

Resource Allocation Modes in C-V2X: From LTE-V2X to 5G-V2X

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Abstract—The paradigm of Internet of Vehicles (IoV) as an extension to the Internet of Things (IoT) concept can assist the development of smart cities. IoV relies on vehicular communications to allow vehicles to communicate in real time with other vehicles, roadside infrastructure, and pedestrians. The main goal is to enable not only road safety services but also time-constrained IoT applications. In this context, Cellular Vehicle-to-Everything (C-V2X) communication is a prominent technology to accomplish IoV goals. Currently, there are two versions of C-V2X available, long-term evolution-V2X (LTE-V2X) and new radio-V2X (NR-V2X), and in both, mechanisms of resource allocation (RA) play an important role in their performance. Therefore, our goal in this article is to present an extensive state of the art of RA on C-V2X technology. Thus, we first present the main technical aspect related to LTE-V2X and NR-V2X technologies. Afterward, we present and categorize the related works for both technologies, pointing out their main differences, advantages, and drawbacks. Finally, we present the challenges, open issues, and future technical trends related to RA in C-V2X.

Index Terms—Cellular Vehicle to Everything (C-V2X), long-term evolution-V2X (LTE-V2X), new radio-V2X (NR-V2X), resource allocation (RA) modes.

I. INTRODUCTION

NOWADAYS, vehicular networks form a large and important research field, attracting automotive and telecommunication industries. Road safety remains the main objective of the automotive industry. In this context, the integration of wireless communication technologies in this sector has led to the emergence of a new concept known as Vehicle-to-Everything (V2X) communication. V2X communications aim to ensure road safety and traffic management services. Furthermore, they are also able to provide other types of service, such as entertainment applications.

Vehicular communication is a prominent component of the Internet of Things (IoT) [1], contributing to the emergence of a new paradigm, which is the Internet of Vehicles (IoV) [2]. The IoV is an advanced application of the IoT in the intelligent transportation systems. The IoT is the key technology

enabler of the smart cities [3], in which, smart objects can communicate with each other. The main goal of the IoV is to enable vehicles to communicate in real time with other vehicles, roadside infrastructure, and pedestrians.

Although its great potential for leveraging smart cities, vehicular communications still face several technical challenges mostly related to its performance and security. In this context, two major technologies have appeared to support vehicular communications and overcome its challenges: 1) Wi-Fi and 2) cellular networks. The Wi-Fi technology, known as 802.11p, operates in the 5.9-GHz band [4]. The cellular technology, known as cellular-V2X (C-V2X) [5], operates not only in the 5.9-GHz band but also in the licensed bands of the cellular networks. The long-term evolution-V2X (LTE-V2X) is the first C-V2X technology standardized by the Third Generation Partnership Project (3GPP) in Release 14. More recently, the 3GPP has also standardized the new radio-V2X (NR-V2X) in Release 16. The IEEE 802.11p performance results indicate its limited performance guarantees, especially for time-constrained IoV applications. However, C-V2X can provide better performances than 802.11p in terms of coverage range, latency, and throughput. Even though C-V2X is a prominent technology to cope with those issues in vehicular networks, solutions based on this technology must deal with resource allocation (RA) problems.

In this regard, the RA in C-V2X has drawn the attention of the research community, in the last few years. In fact, radio frequencies, which are scarce and precious resources, must be efficiently managed to avoid interference and collisions. Besides, vehicular communications impose more challenges to the RA problem due to constant changes in the network topology caused by the high mobility. Therefore, RA in C-V2X is a key aspect to improve its performance and enable IoV applications. As there exist always areas that cannot be served by a base station, C-V2X presents two modes of RA: 1) a centralized undercoverage mode and 2) a distributed out-of-coverage mode. In LTE-V2X, they are, respectively, known as modes 3 and 4. In NR-V2X, they are named modes 1 and 2, respectively. In the undercoverage mode, the evolved Node B (eNB)/generation Node B (gNB) schedules and allocates resources to vehicles. However, in the out-of-coverage mode, the vehicles autonomously select their radio resources using the sensing-based semi-persistent scheduling (SPS) algorithm. Several solutions have already been proposed for the LTE-V2X modes, and especially for mode 4. This is explained by the fact that mode 3 is less challenging thanks to the control of

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TABLE I
QOS REQUIREMENTS OF ADVANCED V2X APPLICATIONS

Use case	Max Latency (msec)	Payload size (Bytes)	Reliability (%)	Data Rate (Mbps)
1. Vehicle platooning: cooperative driving with a group of vehicles dynamically form a group and traveling very closely together, safely at high speed. The platoon leader exchanges periodically information with the other platoon member vehicles	10-500	500-6000	90-99.99	50-65
2. Advanced Driving: aims to enable safer traveling and collision avoidance. Vehicle shares its driving intention and the data obtained from its local sensors with nearby vehicles.	3-100	300-12000	90-99.99	10-50
3. Extended Sensors: with extended sensors, vehicles exchange data with other vehicles through local sensors, which enable vehicles to better perceive their surroundings.	3-100	1600	90-99.99	10-1000
4. Remote Driving : enables the operation of a remotely controlled vehicle. This is useful, for example, when the vehicle is in a dangerous area.	5	-	90-99.99	UL:25, DL:1

the eNB while mode 4 presents different issues related to collision, interference, and congestion control problems caused by its decentralized nature.

Therefore, our goal in this article is to present an extensive state of the art of RA on C-V2X technology. We first present the main technical aspects related to LTE-V2X and NR-V2X technologies. Afterward, we present and categorize the related works for both technologies, pointing out their main differences, advantages, and drawbacks.

To the best of our knowledge, there are many survey papers interested in the RA in vehicular networks. Among them, we cite the recently published survey papers presented in [6]–[9]. Noor-A-Rahim *et al.* [7] presented a state of the art on RA in vehicular networks in general, taking into account both the dedicated short-range communication (DSRC) and C-V2X technologies. However, the RA in C-V2X technology presented in this work is not very in detail and the authors do not cover all the proposed RA algorithms for both in-coverage and out-of-coverage modes, i.e., modes 3 and 4 in LTE-V2X and also modes 1 and 2 in NR-V2X. However, Masmoudi *et al.* [6] and Allouch *et al.* [8] focused on the RA in C-V2X technology, but the limitation of their works is that the authors only focus on the LTE-V2X technology, and so they do not cover the more recent research topics related to the RA in NR-V2X technology. The most recent published survey paper is presented in [9], where the authors present the different RA schemes that can be adopted in a 5G-V2X scenario. However, this article does not present a comprehensive literature review of the already proposed RA algorithms for C-V2X communications.

To fill this gap, we provide a comprehensive overview of RA in both LTE-V2X and NR-V2X technologies. We contribute, in the present work, by presenting the state of the art of the research works regarding modes 3 and 4 of LTE-V2X. Then, we present also the recent works and 3GPP discussions related to modes 1 and 2 of NR-V2X. This article is hence the first survey paper on RA mechanisms that presents both LTE-V2X and NR-V2X technologies.

The remainder of this article is organized as follows. In Section II, we present an overview of communication technologies in vehicular networks, i.e., 802.11p, 802.11bd, LTE-V2X, and NR-V2X. We present also a brief comparison of these technologies and some proposed solutions for

the coexistence of these technologies in the 5.9-GHz band. Section III presents the sidelink RA in LTE-V2X with a detailed state of the art on modes 3 and 4. Last but not least, we investigate in Section IV the sidelink RA in NR-V2X with the most recent proposed RA algorithms for modes 1 and 2 of NR-V2X. In Section V, we present the challenges and open issues related to RA in C-V2X communications. Then, we present the future technical trends and possible solutions in Section VI. Finally, we conclude this article by a conclusion and perspectives in Section VII, in which, we present the future interesting research topics related to the C-V2X RA subject.

II. OVERVIEW ON VEHICULAR COMMUNICATION TECHNOLOGIES

Before getting to the heart of the matter, which is the RA in C-V2X, we start, in this section, by presenting an overview of communication technologies in vehicular networks. The first standardized technology for vehicular networks is the Wi-Fi technology, named 802.11p, followed by the cellular technology LTE-V2X. Today, applications in vehicular networks are evolving with high reliability and low-latency requirements. In Table I, we present the Quality-of-Service (QoS) requirements of advanced V2X applications [10]. The IEEE 802.11p and the LTE-V2X cannot really meet the requirements of future applications. For this reason, standardization bodies have seen the need to evolve toward new technologies. The IEEE created a new Study Group known as IEEE 802.11 Next Generation V2X in March 2018 [11]. This led to the creation of a new group in January 2019, which is the IEEE Task Group 802.11bd [12]. Furthermore, the 3GPP, on its side, has also standardized NR-V2X in its Release 16 [13], which is based on the new radio (NR) technology defined for 5G in Release 15 [14]. These two technologies, i.e., 802.11bd and NR-V2X, are the future technologies for vehicular networks. In the following sections, we begin by a brief presentation of each technology, separately, i.e., 802.11p, 802.11bd, LTE-V2X, and NR-V2X. Then, we present the research works interested in the comparison between these technologies and the potential solutions for the coexistence of these technologies at the 5.9-GHz band.

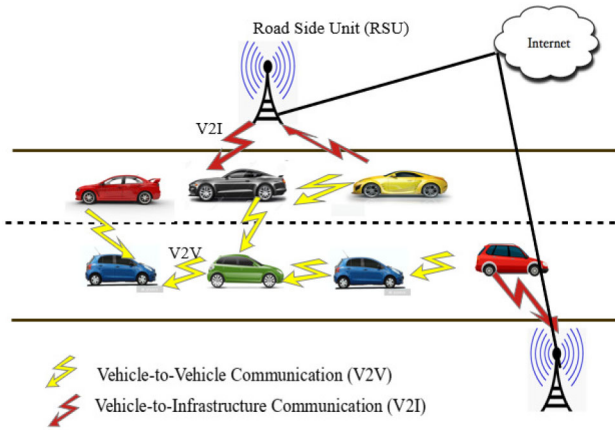


Fig. 1. IEEE 802.11p architecture.

A. Wi-Fi Technologies

1) *802.11p*: The first standardized technology for vehicular ad hoc networks (VANETs), which is now a mature technology, is the ITS-G5 in Europe or DSRC in the USA. This standard is based on the 802.11p protocol [4], a variant of the 802.11a adapted to the dynamic nature of the vehicular environment. It allows two communication types, as presented in Fig. 1: 1) Vehicle-to-Vehicle (V2V) communication and 2) Vehicle-to-Infrastructure (V2I) communication. The infrastructure of a vehicular network refers to the roadside units (RSUs) deployed along the streets and highways.

The 802.11p protocol uses orthogonal frequency-division multiplexing (OFDM) with a 10-MHz channel bandwidth and 8 modulation and coding schemes (MCS) at the physical layer. The maximum bit rate offered by 802.11p is 27 Mb/s. In order to adapt to the dynamic nature of the vehicular environment, the spacing between subcarriers is reduced compared to 802.11a. In the later one, the subcarrier spacing (SCS) is 312.5 kHz, whereas in 802.11p, it is 156.25 kHz, which reduces the Doppler effect. At the medium access control (MAC) layer, the 802.11p adopts the enhanced distributed channel access (EDCA) protocol which adds the QoS to the carrier-sense multiple access with collision avoidance (CSMA/CA) standard. However, if the data load on the wireless channels exceeds the total available capacity, it is necessary here to introduce the decentralized congestion control (DCC) mechanisms in the ITS-G5 stations, as described in [15], to control the channel load. These mechanisms have been designed to meet the requirements of ITS applications on high reliability and low latency, especially for road safety applications.

2) *Toward 802.11bd*: to meet the requirements of future vehicular applications, and thanks to the evolution of Wi-Fi standards from 802.11a to 802.11ax [16], the IEEE has seen the need to standardize a new protocol dedicated for V2X communications. This new protocol, named 802.11bd, adds enhancements to the MAC and physical layers of its predecessor, 802.11p. These new mechanisms allow 802.11bd to achieve a data rate and a communication range two times better than those achieved by the 802.11p protocol in high mobility

TABLE II
COMPARISON BETWEEN 802.11p AND 802.11bd

FEATURES	802.11p	802.11bd
supported relative speed	252 kmph	500 kmph
sub-carrier spacing	156.25 kHz	78.125, 156.25, 312.5 kHz
Channel coding	BCC	LDPC
Midambles	No	Yes
Radio bands	5.9 GHz	5.9 GHz, 60 GHz

scenario with the vehicle's speed reaching 500 km/h. The 802.11bd protocol is also interoperable and backward compatible with 802.11p. It deploys, at the physical layer [12], the following technologies.

- 1) High MCS level up to 256 quadrature amplitude modulation (QAM).
- 2) Multiple-input-multiple-output (MIMO) technology.
- 3) Channel bandwidth up to 20 MHz.
- 4) Dual carrier modulation (DCM) introduced at Wi-Fi 6, which consists of transmitting the same symbol on two different subcarriers.
- 5) Low-density parity check (LDPC) instead of the binary convolutional code (BCC).
- 6) Using midambles instead of preambles for a better channel estimation because midambles are scattered throughout the frame between OFDM symbols.

Table II shows a comparison between the 802.11p and the 802.11bd standards.

B. Cellular Technologies: C-V2X

1) *LTE-V2X*: Recently, 3GPP has standardized in its Release 14 LTE-V2X as a new technology competing with the IEEE 802.11p standard [17]. The first benefit of C-V2X is the availability of the cellular infrastructure, unlike DSRC which requires the deployment of a new infrastructure, i.e., RSUs. LTE-V2X is based on the Device-to-Device (D2D) communications technology [18]. The side-link interface, known as the PC5 interface, has been standardized for D2D communications. Enhancements are added to this interface in order to support the high mobility of the vehicular environment, especially to support the high speeds of vehicles. Thus, new demodulation reference symbols (DMRS) are added to the frame structure to handle the Doppler effect.

While the 802.11p protocol only allows V2V and V2I communication, C-V2X provides the integration of both direct communication, i.e., V2V and V2I, based on the Sidelink PC5 interface, and cellular communication, i.e., Vehicle to Network (V2N), based on the Uu interface. In order to support V2X communications, two main elements are added to the LTE core network architecture, the V2X control function and the V2X application server [19], as illustrated in Fig. 2. The V2X control function provides the user equipment (UE) with the necessary parameters for V2X communications, and the V2X application server is responsible for the V2X communication data treatment. In LTE-V2X, two communication modes are possible: mode 3 and mode 4. The eNB schedules the radio resources and allocates them to vehicles in mode 3. However, in mode 4, vehicles reserve their radio

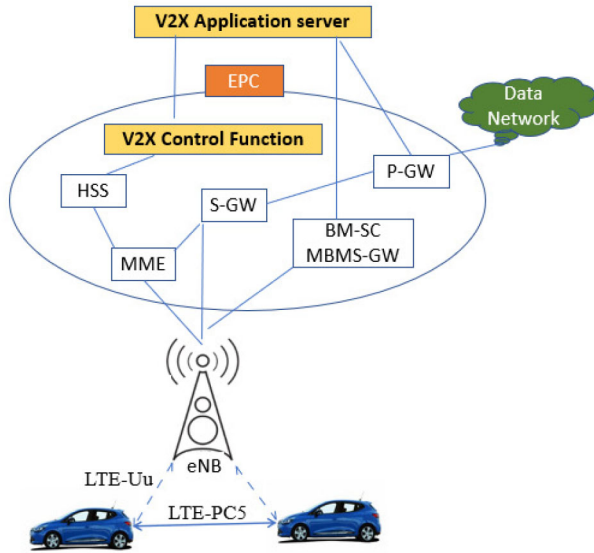


Fig. 2. LTE-V2X architecture.

resources autonomously using the sensing-based SPS algorithm [20]. It is worth noticing that mode 3 depends on the cellular network to work properly, while mode 4 can operate without any cellular network.

2) *5G-V2X/NR-V2X*: The NR-V2X was designed to support advanced V2X applications [21]. Vehicle platooning, extended sensors, advanced and remote driving are among these applications. Such applications require low latency and high reliability. NR-V2X supports these new applications as well as those related to the basic safety service already supported by LTE-V2X. In case a vehicle is equipped with the two radio access technologies (RATs) of C-V2X technologies, i.e., LTE-V2X and NR-V2X RATs, it can use the LTE-V2X RAT for basic safety applications and NR-V2X RAT for these new applications. Depending on the application, the exchanged messages can be of periodic or nonperiodic nature. Furthermore, some messages are broadcasted for all vehicles while others are targeted to a specific group of vehicles, such as in the platooning scenario, where the platoon leader communicates with the platoon members. For this purpose, the NR-V2X supports the unicast and groupcast modes besides the broadcast mode, already supported in LTE-V2X [22]. These new modes can operate under and outside the cellular coverage. Note that the vehicle can use both groupcast and broadcast modes at the same time, i.e., the vehicle can communicate with a group of vehicles in the groupcast mode, and simultaneously send its messages to other vehicles in the broadcast mode. The platooning is the best example for this case, where the platoon leader communicates with platoon members in the groupcast mode to maintain the platoon and also broadcasts the periodic message to other vehicles outside the platoon for the cooperative awareness service.

a) *NR-V2X key features*: NR-V2X has the following key features.

- 1) *Support for New Numerologies*: Indeed, in 3GPP Release 15, new numerologies have been standardized

TABLE III
COMPARISON BETWEEN LTE-V2X AND NR-V2X

FEATURES	LTE-V2X	NR-V2X
sub-carrier spacing	15kHz	15, 30, 60, 120 kHz
Communications modes	Broadcast	Unicast, Groupcast, Broadcast
MCS	QPSK, 16QAM	QPSK, 16QAM, 64QAM
Waveform	SC-FDMA	OFDM
PSCCH and PSSCH	FDM	TDM
Feedback Channel	No	PSFCH
DMRS /sub-frame	4	Flexible
Sidelink modes	3, 4	1, 2

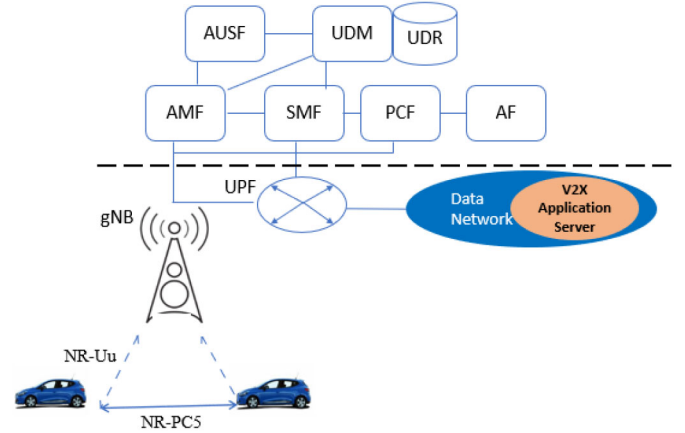


Fig. 3. 5G system architecture for V2X communications.

for NR of 5G [23]. These numerologies are also supported by NR-V2X standardized in Release 16. Unlike LTE-V2X, where the spacing between subcarriers is fixed on 15 kHz, in NR-V2X, it can take other values multiple of 15 kHz, i.e., 30, 60, and 120 kHz. As the SCS is variable, the time slot, i.e., the time to transmit 14 OFDM symbols, is also variable and decreases with the increase of the SCS, which reduces the latency and therefore favors critical applications in terms of latency. A more detailed comparison between LTE-V2X and NR-V2X is presented in Table III.

- 2) *Multiplexing of PSCCH and PSSCH in Time Domain*: Another key feature of NR-V2X, which also favors the reduction of latency time is the multiplexing of physical sidelink control channel (PSCCH) and physical sidelink shared channel (PSSCH) in the time domain, i.e., the PSCCH will be transmitted first, followed by the transmission of PSSCH, whereas in LTE-V2X, these channels are multiplexed in the frequency domain.
- 3) *Use of the Physical Sidelink Feedback Channel (PSFCH)*: This new channel has been defined in NR-V2X in order to ensure the reliability for unicast and groupcast modes.

NR-V2X also provides other enhancements on its physical layer, such as the high MCS level up to 64 QAM, the LDPC coding, and a flexible number of DMRS symbols in the slot.

As in LTE-V2X, the 5G system architecture includes the V2X applications server as illustrated in Fig. 3, which is responsible for the V2X communication data treatment. However, the necessary parameters for the establishment of

V2X communication are provided by the policy control function (PCF) unit to the UE while those parameters are provided by the V2X control function in LTE-V2X.

C. Comparison Between Cellular and Wi-Fi Technologies

Nowadays, telecommunication operators are asking for the abandonment of the ITS-G5 and the support of the C-V2X as the best technology for vehicular communications. In the USA, Japan, and South Korea, projects for the deployment of the 802.11p are already in place. In China, an open road test for LTE-V2X was carried out in September 2018. In Europe, the C-Roads Platform [24], deploying the cooperative ITS, is aiming for 6,000 km of ITS-G5 coverage and 200,000 km of cellular coverage. On the global scale, the 5G Automotive Association (5GAA), which is the first supporter for the C-V2X, is preparing for the integration of 5G in the automotive sector.

Given this technological competition and diversity of choice, much research work have focused on the comparison between the performance of these two technologies [25]–[33]. Research in this subject has proved that both technologies present some advantages and drawbacks, which depend on many factors, such as the line-of-sight (LOS) conditions, the vehicular density, the size of transmitted packets, and also the distance between transmitter and receiver.

On the other hand, a special attention is given to the comparison between 802.11p and LTE-V2X mode 4 as they are two decentralized technologies operating without a fixed infrastructure. Cecchini *et al.* [25] made a comparison between LTE-V2V mode 3, mode 4, and 802.11p. They demonstrate that the LTE-V2V mode 3 outperforms both the LTE-V2V mode 4 and the 802.11p standard. However, the comparison between 802.11p and the LTE-V2V mode 4 depends on the distance between transmitter and receiver. For smaller distances, 802.11p outperforms the LTE-V2V mode 4, while for longer distances, the inverse happens.

It is important to point out that despite the differences between these two technologies, some schemes can work in both technologies. One example is positioning methods used to estimate the positions of neighboring vehicles, which is particularly interesting for safety applications. Indeed, the accuracy of the positions of vehicles and the number of neighboring vehicles in a given communication range is interesting since it can be used in the design of the RA algorithm. Moreover, these position-based methods can be used in both DSRC and C-V2X technologies. In this context, Cai *et al.* [34] addressed the problem of estimating neighbor positions in VANETs. The authors propose a new efficient distributed method for estimating the number of neighboring vehicles and their distribution, called PeerProbe, in which neighboring vehicles cooperate with each other to explore the neighborhood. PeerProbe is validated by simulations considering different channel conditions and vehicle densities. The authors demonstrate that PeerProbe achieves better performance in terms of vehicle positioning and distribution accuracy.

Other research works have proposed a hybrid solution by exploiting the benefits of both technologies in order to support

an efficient and reliable V2X communications. In this context, Abboud *et al.* [35] investigated the potential mechanisms for the coexistence of DSRC and C-V2X. As proposed in this work, when both technologies are used, the cellular network can have two functions. The first one consists of using it as a backup for vehicular data in case the V2V connections are interrupted. In the second one, the cellular network can serve, for example, as an access network to the Internet. In the same context, King *et al.* [36] addressed the interoperability issue between these two technologies. The authors propose a converter protocol for the translation between these two technologies. In the first step of this protocol, each vehicle broadcasts a message in order to identify the used technology by all its neighbors and then saves this information in a knowledge table. The authors are based on a cyclostationary analysis for the technology identification. In fact, generally, each type of wireless signal has unique cyclic frequencies with the frequency selective property, which distinguishes itself from other types of signals, and the cyclostationary analysis can be used to identify the type of the signal. Thus, whenever a vehicle wants to transmit a message it must access its knowledge table to know whether the destination vehicle uses the same technology or not. When both source and destination use the same technology, the message is generated and transmitted as usual. However, if it is not the case, the proposed converter protocol is used to convert the signal into the necessary format of the other technology.

Here, we highlight that there are other recent research works interested in the issue of data hungry V2X applications. As an example of those kinds of applications, we consider the see through (ST) application [37], in which a video stream is transmitted between vehicles in order to enhance the overtaking decision between them. Recently, some research works propose the use of new techniques to enhance the visibility between vehicles. In this context, Masini *et al.* [38] discussed the use of reconfigurable metasurfaces in vehicular networks. The reconfigurable metasurfaces can be used so that the environment, such as buildings, vehicles, and more others, can act as an active part of the communication between vehicles. The use of those reconfigurable metasurfaces enables the environment to be smarter and then enhances the reliability of V2X communications by enhancing the driver's visibility. The authors of this article discuss the impact of the use of those metasurfaces on the basic use cases of V2X applications, such as the collision avoidance and the vulnerable road users (VRUs) detection. The authors demonstrate that the use of metasurfaces enables to extend the network coverage and also it does not affect the network congestion level.

D. Coexistence Between Cellular and Wi-Fi Technologies

The coexistence between cellular and Wi-Fi technologies is a very relevant research subject. We can categorize two types of coexistence. The first one is the intertechnology coexistence that we refer to as the coexistence between cellular and Wi-Fi technologies, i.e., between LTE-V2X/NR-V2X and 802.11p/802.11bd. The second type is the intratechnology

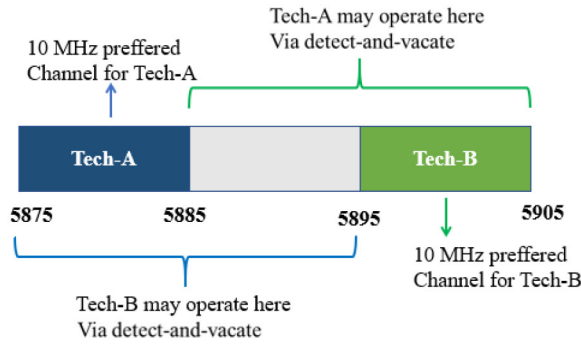


Fig. 4. 5GAA model for the coexistence between IEEE 802.11p and C-V2X in the 5.9-GHz band.

coexistence that we refer to as the coexistence between LTE-V2X and NR-V2X.

1) *Intertechnology Coexistence*: The coexistence between cellular technologies, i.e., LTE-V2X or NR-V2X and the Wi-Fi technologies, i.e., 802.11p or 802.11bd in the 5.9-GHz band, is a topic of interest as discussed in [39]. As a proposal for the coexistence in the same band, the 5GAA has proposed a solution based on the division of the spectrum between these technologies in order to avoid the co-channel interference problem. In [40], it is proposed to dedicate to each technology a 10-MHz channel in the ITS band, and then each technology can also operate in the rest of the ITS spectrum by using a specific detect-and-vacate mechanism, as presented in Fig. 4.

2) *Intratechnology Coexistence*: NR-V2X is not backward compatible with LTE-V2X, because NR-V2X uses different numerologies. An LTE-V2X vehicle with a carrier spacing of 15 kHz is not able to decode the message of an NR-V2X vehicle using a carrier spacing of 30 kHz. Future vehicles can probably be equipped with both RATs, i.e., LTE-V2X and NR-V2X. These two technologies can operate in the same band, i.e., 5.9-GHz band, but in two different channels. Two main approaches of coexistence are proposed in [41].

a) *Coexistence with FDM*: In this approach, LTE-V2X and NR-V2X simultaneously operate on two different channels. The drawback of this approach is that if the two RATs are using the same band, the total permitted transmission power must be therefore divided between them. The authors of [42] propose that the division can take into account the priority of packets by increasing the transmission power of the RAT with the higher priority packet at the expense of the transmission power of the second RAT.

b) *Coexistence with TDM*: This second approach considers that both technologies must operate in orthogonal channels in time and frequency domains. In contrast to the first one, it allows both RATs to use the maximum permitted transmission power because they will not transmit simultaneously. However, this approach is not practical for the transmission of packets of latency-critical applications, because the packet has to wait during the transmission time of the second RAT, which can increase the latency time. On the other hand, in the context of intratechnology coexistence within WiFi standards, Khan and Härrä [43] proposed two protocols: 1) detect and

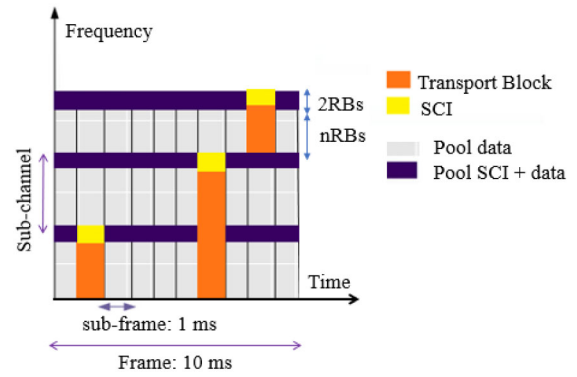


Fig. 5. LTE-V2X frame structure.

mitigate (DAM) and 2) detect and vacate (DAV), as solutions for the co-channel coexistence.

Next, in the two following sections, we present the sidelink RA in LTE-V2X and NR-V2X, respectively.

III. SIDELINK RESOURCE ALLOCATION IN LTE-V2X

Here, we explain the general concepts related to the resources in LTE-V2X in the first section. The second section presents, in-depth, the details of the RA modes in LTE-V2X, i.e., modes 3 and 4. Finally, in the third section, we investigate the state of the art of each mode, separately.

A. Resources in LTE-V2X

As in LTE uplink, LTE-V2X adopts OFDM in its physical layer and the single-carrier frequency-division multiple access (SC-FDMA) at the MAC layer. The available bandwidth, i.e., a 10- or 20-MHz channel, is subdivided in time and frequency domains into several orthogonal resources, as illustrated in Fig. 5. In the time domain, the signal is organized into frames of 10 ms. Thus, each frame is formed by ten subframes of 1 ms and each subframe is formed by two time slots (TS). In the frequency domain, the signal is formed by resource block pairs (RBP), which is defined by 12 subcarriers, spaced by 15 kHz, and carrying 14 OFDM symbols. In LTE-V2X, a subchannel is defined by a group of RBPs in the same subframe. The subchannel size can be configured from 4 to 50 RBPs as defined in [44] but all subchannels in an eNB must have the same subchannel size. Each vehicle can use one or multiple subchannels to transmit its data. A subchannel is defined as the smallest unit of resource that can be assigned to a vehicle to transmit its CAM packet. In C-V2X, quadrature phase-shift keying (QPSK) and 16 QAM modulations are used to transmit the transport block (TB), whereas the sidelink control information (SCI) is always transmitted using QPSK. LTE-V2X uses the Turbo coding and the normal cyclic prefix (CP). The transport channel (TCH) transporting data is carried by the PSSCH while the PSCCH carries the SCI. The SCI provides the critical informations in order to successfully decode the message at the receiver, such as the MCS information, the RBs occupied by the associated TB, the priority of the message, and the resource reservation interval (RRI). The RRI enables to determine when the vehicle will use the reserved resources

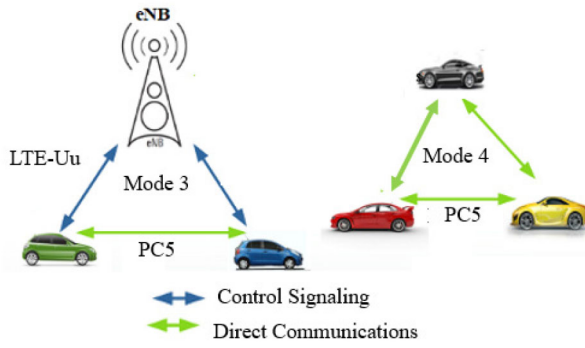


Fig. 6. RA modes in LTE-V2X.

for its next transmission. PSCCH and PSSCH are multiplexed in frequency domain, i.e., they are transmitted in the same subframe but using different frequency resources. There are two configuration schemes for the PSSCH and PSCCH. The first one is the adjacent configuration, where the PSCCH occupies the two first RBs of the allocated subchannel, and it is directly followed by the PSSCH. The second configuration is a non-adjacent configuration, where the PSSCH and the PSCCH do not occupy successive RBs in the same subframe. The number of RBs in a subchannel is variable [45]. Together with this information, are also sent four DMRS symbols inserted in the OFDM subframe, which are used for channel estimation in high mobility environments.

B. Sidelink Resource Allocation Modes in LTE-V2X

LTE-V2X presents two modes of RA as illustrated in Fig. 6. These modes are related to the availability of the cellular infrastructure. The first mode is applied under the cellular coverage and is known as mode 3. In this mode, the eNB schedules and allocates resources to vehicles. However, in order to ensure the basic safety services even though the vehicles are out-of-the-cellular coverage, the 3GPP has standardized a second mode known as mode 4. In mode 4, vehicles autonomously select their radio resources using the sensing-based SPS algorithm. The SPS is based on the sensing of the channel before choosing a resource from a list of candidate resources, as we will detail later.

1) *Mode 3*: In contrast to mode 4, 3GPP has not standardized a specific algorithm for RA for mode 3. Therefore, each operator can propose its own algorithm. However, there are two choices for RA in this mode.

a) *Dynamic allocation*: In the dynamic allocation, each vehicle requests subchannels to the eNB for each CAM packet transmission. The drawback of the dynamic allocation is the increase of the signaling overhead and the latency.

b) *Semi-persistent scheduling*: In the SPS, the eNB reserves subchannels for the periodic transmissions of vehicles. Each vehicle sends to the eNB the necessary information such as packet size, transmission frequency, and priority of packets. This information is referred to as the assistance information, which is used by the eNB to semi-persistently reserve the appropriate resources for each vehicle. The eNB can configure the vehicle with up to eight SPS

configurations [46]. For each configuration, a specific index and a different periodicity of the sidelink transmission are specified. The SPS configuration is used by a vehicle just if the eNB indicates to the vehicle that this configuration is active. The main advantage of the SPS approach is the reduction of the signaling overhead and the delay of the RA mechanism.

It is important here to note that the periodic nature of the basic safety messages, on which are based the primary use cases in LTE-V2X, makes the semi-persistent-scheduling approach the more suitable for the V2V services.

2) *Mode 4*: Mode 4 is the mode standardized by 3GPP for the autonomous radio resource selection. In this mode, vehicles autonomously select their radio resources using the sensing-based SPS algorithm, which is more detailed in [47]–[49]. In the following, we present the sensing-based SPS algorithm.

a) *Sensing-based semi-persistent scheduling algorithm*: The SPS is based on the sensing within a (pre)configured resource pool before choosing a resource from a list of candidate resources. The basic principles are as follows.

- 1) When a vehicle reserves one resource, this resource is used to transmit a random number of consecutive messages. This random number, also called the reselection counter, depends on the periodicity of the CAM messages. The reselection counter is chosen between [5–15] for a periodicity of 10 Hz, [10–30] for a periodicity of 20 Hz, and [25–75] for a periodicity of 50 Hz.
- 2) The periodicity of the CAM messages and the value of the reselection counter are included in the SCI field. Thus, based on this information, the vehicle can know which resources are free and which ones are occupied by other vehicles.
- 3) After each transmission of a CAM message, the value of the reselection counter is decremented by 1. When the reselection counter reaches zero, a new resource must be selected with a probability of $(1 - P)$, where P is the probability of keeping the same resource for the next transmissions.

We can divide the operation of the SPS algorithm in three main steps, as illustrated in Fig. 7.

In the first step, the vehicle reserves candidate resources within a selection window (SW). The SW is a time window defined by $[T + T_1, T + T_2]$, where T is the starting time of the subframe at which the vehicle wants to select a new candidate resource. T_1 is the processing time (in subframes) required by a vehicle to identify and select candidate resources for transmission with $1 \leq T_1 \leq 4$. T_2 is also selected by the vehicle and must be included within the range $20 \leq T_2 \leq 100$. Then, the vehicle listens to the channel during the “Sensing Window” for a period of 1 s. The Sensing Window corresponds to the last 1,000 subframes. In Fig. 8, we illustrate the sensing and SWs for the SPS algorithm.

In the second step, the vehicle creates a first list L_1 containing the resources previously selected during the SW, except those presenting a level of reference signal received power (RSRP) higher than a fixed power threshold Th . Also, L_1 excludes the resources occupied by other vehicles for their next transmissions according to the SCI information. This list

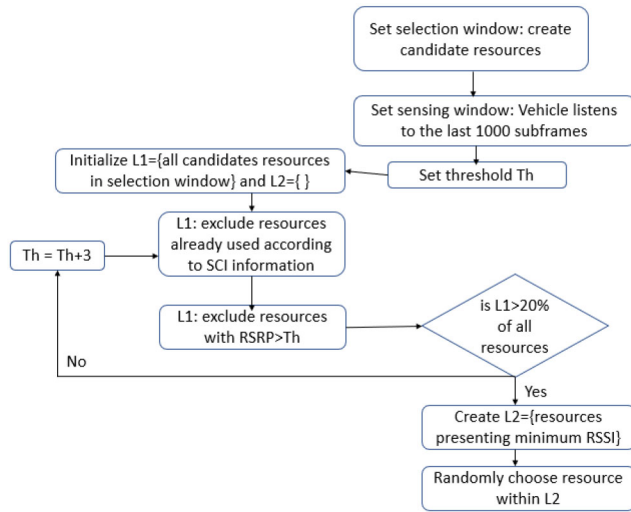


Fig. 7. Sensing-based SPS algorithm for mode 4 in LTE-V2X.

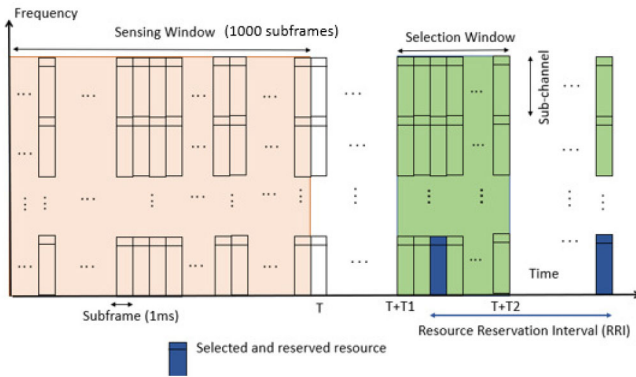


Fig. 8. Sensing and SWs in LTE-V2X.

must contain at least 20% of the total resources selected in the first step. Otherwise, this step is iteratively repeated by increasing the power threshold Th each time by 3 dB.

In the third step, the vehicle creates a list of resources L_2 presenting the minimum RSSI values. The number of these resources must be equal to 20% of the total resources selected in the first step. Thus, the vehicle chooses randomly a final resource from the L_2 list and the chosen resource is reserved by the vehicle for its next transmissions according to its reselection counter parameter. Random selection avoids collision problems in case two vehicles select the same resource presenting the lowest RSSI value.

LTE-V2X mode 4 can also support the allocation of resources based on the geographical position of zones, where a geographical area can be divided into several zones. Different resource pools are assigned to nearby zones, with the application of a distance-based spatially resource reuse approach.

It is worth mentioning that there can be performance degradation in LTE V2X mode 4 due to packet collision problems with higher channel load. In this context, the 3GPP has not standardized a specific congestion control mechanism, but it has specified the following two metrics for determining the channel congestion.

- 1) *Channel Occupancy Ratio (CR)*: It provides an indication on the channel occupancy generated by the transmitting vehicle. It is calculated at the subframe n as the ratio between the total number of subchannels used by the vehicle in subframes $[n - a, n - 1]$ and reserved by the vehicle for its next transmissions in subframes $[n, n + b]$ and the total number of subchannels within $[n - a, n + b]$. Parameters a and b must satisfy $a + b + 1 = 1000$ with $a \geq 500$.
- 2) *Channel Busy Ratio (CBR)*: It is the ratio between the number of subchannels presenting RSSI values higher than a fixed threshold and the total number of subchannels in the last 100 subframes. The CBR provides an estimation on the state of the channel.

The LTE-V2X standard defines up to 16 CBR ranges and it defines for each one a fixed CR_{Limit} that cannot be surpassed by a vehicle. Thus, before the vehicle sends a message, it first measures the CBR and gets the associated CR_{Limit} . Then, the vehicle estimates its appropriate CR. When the measured CR exceeds the CR_{Limit} , the vehicle must adjust its transmission parameters. To this aim, the vehicle can, for example, discard certain packets or augment the MCS to use less subchannels and hence reduce its CR.

3) *Mode 3 and Mode 4 Resource Pool Sharing*: In fact, the radio resources are (pre)configured separately for vehicles operating in mode 3 and those operating in mode 4. In LTE-V2X Rel-14, the SCI field transmitted by a vehicle operating in mode 3 does not include the SPS reservation period of the reserved resource. As a result, a vehicle operating in mode 4 does not exclude the occupied resources by vehicles operating in mode 3 during the resource selection procedure of the SPS algorithm. This can lead to the collision problems between vehicles operating in mode 3 and those operating in mode 4. However, to solve this problem, the mode 3 SCI field in Rel-15 indicates the SPS reservation period of the reserved resource. Therefore, vehicles operating in mode 4 of both releases can exclude the occupied resources by vehicles in mode 3 during the SPS algorithm [50].

C. State of the Art on Modes 3 and 4

Here, it is worth mentioning that mode 4 has a huge volume of research works in the literature compared to mode 3. This is explained by the fact that mode 3 is less challenging thanks to the control of the eNB, while mode 4 presents different issues related to collision, interference and congestion control problems caused by its decentralized nature. However, even though mode 3 has to cope with a significant reduced set of problems, it remains very important to propose new RA algorithms aiming to optimize the network capacity, the data rate of vehicles or the reuse of resources between vehicles.

In the following, we present the state of the art of RA in mode 3 and mode 4, respectively.

1) *State of the Art on Mode 3*: Abanto-Leon *et al.* [51], proposed a new RA solution in order to guarantee the QoS for each vehicle. In this context, the QoS refers to the number of subchannels and the data rate that each vehicle needs for its transmission. This proposal also aims to avoid the main

conflicts that might occur between vehicles during transmissions. These conflicts are mainly related to the intracluster and interclusters interference problems, assuming that vehicles are grouped into clusters based on their geographical location. The intracluster problem is due to the half-duplex (HD) nature of the sidelink interface PC5. In fact, two vehicles in the V2V range cannot transmit simultaneously at the same time. The interclusters problem is related to the positioning of vehicles at the intersection of clusters. Those vehicles can receive messages broadcasted by other vehicles belonging to different clusters using the same subchannel simultaneously, which cause interference problems. The authors use a mathematical framework in order to propose an RA solution that can avoid these conflicts. However, since this proposal is based on the vehicle clustering approach, the stability of the cluster and the periodicity of the clustering algorithm is a challenging issue due to the high mobility of vehicles. The authors of this article do not provide detailed information about the used clustering approach. In any case, we note that a poor clustering approach may affect the performance of the proposed RA scheme. Then, the same authors have represented the centralized RA in mode 3 by a weighted bipartite graph [52]–[54]. In these proposed solutions, vehicles and spectral resources are represented by the vertices of a graph and the edges represent the achievable data rate in each resource. In [52], the RA solution aims to efficiently manage resources by avoiding the conflicts previously cited in [51] and also by enabling the reuse of resources between vehicles with an interference avoidance mechanism. The authors show mathematically and through simulations that this approach improves the performance in terms of the achieved data rate requested by vehicles. Then, in [54], the main objective of the proposed algorithm is to find the optimal one-to-one vertex assignment in order to maximize the capacity of the system defined as the achieved data rate. However, in [53], it is assumed that vehicles monitor the sidelink subchannels with the sensing mechanism and send to the eNB the perceived SINR on these subchannels. The eNB efficiently allocates resources to vehicles based on this information. However, the perceived SINRs are sent by vehicles to eNB via the uplink interface, using a part of the bandwidth. Next, the authors try to quantify the SINR granularity and evaluate the impact of this quantization on the performance of the system. The same authors propose in [55] two RA schemes. The first scheme aims to minimize the perceived power of subchannels. In this approach, it is assumed that vehicles send via uplink interface the channel conditions to the eNB. Then, eNB schedules resources based on this information with a minimization of perceived power of allocated subchannels in order to allocate suitable subchannels. The suitable subchannels are the nonoccupied subchannels or those subchannels experiencing a negligible interference. However, the second approach aims to maximize the distance for the reuse of subchannels between vehicles in order to avoid the co-channel interference. The authors demonstrate through simulations that these two approaches outperform the decentralized approach, i.e., mode 4, in terms of the packet reception ratio (PRR), because interferences are mitigated in the proposed RA algorithm.

Cecchini *et al.* [56], [57] and Fritzsche and Festag [58] focused on the geographical positions-based RA schemes and the distance-based resource reuse approaches. In [56], the RA solution is based on the knowledge of the position of vehicles. The position of vehicles is known through the uplink time difference of arrival (UTDOA) or the Global Navigation Satellite System (GNSS) positioning. They compare the effect of the positioning error on the system performance. They demonstrate that the accuracy of the localization is more important than the frequency of the GNSS update, which lead to a low appreciated error rate. Besides, the authors propose that a resource can be allocated for two vehicles if the distance between those vehicles is greater than a fixed resource reuse distance. Next, the same authors propose a new algorithm that aims to maximize the resource reuse distance between vehicles in order to control the probability of interference [57]. The authors prove that this last algorithm allows to improve the PRR compared to its predecessor, presented in [56].

The study in [58] does not propose a specific RA scheduling in mode 3, but it demonstrates analytically the existence of an optimal resource reuse distance that maximizes the throughput. This work is also validated by the authors through Monte Carlo simulations.

The limitation of the aforementioned algorithms is that they do not take into account traffic density and channel conditions simultaneously, which is in contrast to the work presented in [59]. Sempere-García [59] proposed to use the geographical location of the vehicles and the environmental conditions to allocate resources to vehicles. This proposal is called adaptive spatial reuse of radio resources (DIRAC). DIRAC defines two distances: 1) the reuse distance and 2) the HD distance. The reuse distance is the distance between two vehicles reusing the same resource and the HD distance is the distance between two vehicles transmitting in the same subframe but using different subchannels. This proposal adapts the computed distances to the context conditions, such as the traffic density and the channel load. DIRAC aims to minimize the probability that two interfering vehicles are close to each other in order to minimize the QoS degradation. The simulation results show that DIRAC outperforms existing LTE-V2X mode 3 and mode 4 scheduling schemes in terms of the packet delivery ratio (PDR) in both LOS and NLOS conditions.

Based on the aforementioned research works, we note that many RA proposals in Mode 3 are based on the geographical positions of vehicles. However, despite the performance shown by these algorithms, there may be performance degradation due to the inaccurate positioning of vehicles. Moreover, we point out another limitation of the proposed RA algorithms, namely, the failure to take into account the QoS requirements of the various V2X applications. Indeed, vehicular applications include not only safety applications but also other types of services, such as traffic management applications and entertainment applications. In this context, a recent research work has investigated the RA considering different QoS requirements of safety-related and nonsafety-related traffic. In this work, Allouch *et al.* [60] proposed a new algorithm for mode 3, named priority and guaranteed-based resource allocation (PEARL) algorithm. The main goal of the

PEARL algorithm is to allocate resources to vehicles considering the QoS requirements of the V2X applications. In the PEARL algorithm, the authors classify the V2X traffic into two types: 1) safety traffic and 2) nonsafety traffic. In this work, the authors aim to guarantee the minimum of resources to the basic safety service. For that, the algorithm reserves a guaranteed group of resources for the safety traffic, which is used in case of high traffic density. The guarantee of those resources aims to ensure that the vehicles have always the needed resources to transmit their basic safety messages. Then, the remaining resources are divided between the safety traffic and the nonsafety traffic. However, in dense traffic scenario and when the reserved resources for the safety traffic are all consumed, the PEARL uses a priority-based RA scheme. In this scheme, a high priority is attributed to the safety messages and a low priority is assigned to nonsafety messages. The simulations demonstrate that the PEARL algorithm can achieve the high reliability and low latency required by the safety traffic without impacting the transmission of the nonsafety messages. Recently, the same authors have extended the PEARL algorithm to a new version called PEARL-RM [61]. In PEARL-RM, a new resource reuse mechanism (RM) is proposed in order to reuse resources between vehicles of both safety and nonsafety traffic.

In addition, recently, Khabaz *et al.* [62] have proposed a new RA algorithm for mode 3 based on the clustering of vehicles. In this work, the authors are interested in the RA for basic safety traffic. This algorithm, named maximum inter-centroids reuse distance (MIRD), aims to avoid the blocking of transmitters caused by the unavailability of radio resources in case of the high vehicle density area. In fact, the blocking of transmitters is very dangerous in the basic safety services because the blocked vehicle is not detected by its neighbors. For that, the authors have proposed an efficient RA algorithm that facilitates the reuse of resource between clusters of vehicles. The main idea of this proposal is first the grouping of vehicles into clusters according to their geographical positions and directions. Then, the eNB divides the available resources into orthogonal resource sets and assigns a resource set to each cluster. In the final step, MIRD applies a specific resource reuse approach between clusters by computing the intercentroids distance between clusters. In this resource reuse approach, two clusters can reuse the same resource set if their intercentroid distance is greater than a fixed threshold.

As previously pointed out, there are no much research works investigating mode 3 in the literature, so this mode remains until now an immature research subject and need more investigative research works.

2) *State of the Art on Mode 4:* We can classify the research works related to mode 4 in three main categories. The first category is interested in the evaluation of the impact of the SPS parameters on the performance of mode 4. The second category demonstrates the need to modify the SPS algorithm by adding new procedures in order to enhance the performance of mode 4. The third category proposes new alternatives to the SPS algorithm and demonstrates that those proposed algorithms outperform this last one.

As previously pointed out, the first category of research works, which group the largest number of research works, is interested in the improvement and the evaluation of the performance of the SPS algorithm. In the literature, the first work presenting an evaluation of the performance of the SPS algorithm is presented in [49]. In this work, the evaluation of the SPS algorithm is done under realistic traffic conditions in urban scenarios. The authors demonstrate through simulations that the SPS algorithm can experience significant performance degradation due to packet collisions. In addition, Nabil *et al.* [63] demonstrated that the RRI significantly influences the packet data rate performance. The authors conclude through simulations that the PDR increases with the increase of the RRI. This is explained by the fact that if this interval increases, the probability that another vehicle uses the same resource decreases, thus, the PDR increases.

However, Molina-Masegosa *et al.* [64] demonstrated that the increase of the probability of reselection P can improve the performance of the SPS algorithm when the vehicles transmit at 10 packets per second (PPS). This is explained by the fact that increasing P leads to a decrease in the number of resource reservations, which enhances the stability of the sensing environment. However, simulations demonstrate that when the channel load increases, the increase of P can reduce the PDR. In fact, the increase of the channel load reduces the communication range, and due to the high mobility of vehicles, when a vehicle selects a resource and keeps it for a long time, it is very likely that it will come within range of vehicles that were not previously in its sensing range and were not considered in the resource selection by the sensing-based SPS scheme. As a result, this vehicle will experience packet collisions, which decreases the performance of the SPS algorithm. In addition, this study shows the advantages of using a low RSRP threshold during the second step of the SPS algorithm and a high transmit power.

The above-mentioned research works are based on simulations to evaluate the performance of the SPS algorithm. However, Gonzalez-Martín [47] used analytical models for this purpose. The authors have chosen the PDR as the performance metric. The study models four main types of errors that can affect the quality of transmission in mode 4. Those errors are mainly related to the HD nature of the sidelink interface PC5, propagation conditions, and collision problems. The study also aims to compare the results obtained with the analytical models to those obtained with the simulator implemented over the Veins platform [65]. This work demonstrates that analytical models are capable of accurately model the performance of the LTE-V2X Mode 4 communication.

Recently, Bartoletti *et al.* [66] studied the effects of the generation of the CAM packet on the overall performance of autonomous mode 4. In this work, the authors study the case where the generation of the CAM packet is not exactly periodical even though the CAM is continuously exchanged. For that, they consider the case where the CAM packet generation interval T_g is larger or shorter than the allocation period T_b . The authors study the impact of those parameters on the overall performance considering the PRR parameter. The simulation results show that the optimal value of T_b depends on

different conditions but a value of 0.1 s is the optimal one. Furthermore, they demonstrate that the overall performance of mode 4 reduces as soon as the CAM packet generation interval is not equal to the allocation period.

The limitation of the first category of research works is that the authors do not investigate the impact of the SPS algorithm of the autonomous mode 4 on the stringent requirements of V2X applications, such as latency and reliability of safety messages under different channel conditions. This aspect needs to be studied more carefully since the autonomous mode 4 is considered as a basic mode due to its independence from the cellular coverage. After the indication that the SPS algorithm might experience significant performance issues, the need for further enhancements in the mode 4 operation was clear. Therefore, the second category of research works is about modifying the SPS algorithm to improve the mode 4 performance.

The collision problem is among the main performance issues detected by previous works since it significantly affects the overall performance. In fact, in the resource selection phase of the SPS algorithm, a packet collision can happen if two or more vehicles select the same radio resource. Even if the random selection in the final step of the SPS algorithm helps to reduce the collisions between vehicles, it cannot totally cope with this problem and the probability of packet collisions remains not negligible.

Hence, Bonjourn *et al.* [67] proposed a new mechanism for the SPS algorithm that can avoid the collision problems. This solution aims to decrease the transmission collision probability and to improve the reliability in mode 4 communication. In fact, in the SPS algorithm, the overlapping reselection window can cause the collision between vehicles. To avoid this problem, the authors propose a counter learning and reselection (CLR) mechanism. In this mechanism, vehicles transmit the counter values in each packet transmission. The vehicles are then informed about future concurrent reselections. In this way, the system will not allow the vehicles to perform resource reselection which can cause a collision within the reselection window.

In addition, Jeon and Kim [68] proposed a new resource reservation solution that aims to avoid the packet collisions. This proposal aims to enhance the SPS algorithm with an additional reservation mechanism. In this reservation mechanism, each vehicle announces the resource reservation information by using a time–frequency coordinate of the reserved resource. This resource reservation is also made much earlier than the actual use of this resource and so it requires a lot of announcements in order to enhance the reliability of this reservation information. The authors demonstrate through simulations that this proposal achieves better performance than the standard SPS in terms of reliability and latency. However, the proposed solution can lead to the overload problem that can be caused by the required announcements of the reserved resource. Another short-term sensing-based resource selection (STS-RS) scheme is proposed by He *et al.* [69]. In this proposal, a short-term sensing is used and configured before the resource selection phase. In this scheme, each transmitting vehicle can start the STS-RS from the beginning of the resource SW defined in

the standard SPS algorithm. Then, the vehicle performs the short-term sensing measurements during a configured back-off timer. As a result, the selection of the appropriate resource depends on the result of this short-term sensing procedure. The simulation results show that the proposed scheme has a better performance in terms of reliability by reducing the packet collisions probability.

On the other hand, the resource selection phase in the SPS algorithm is based on the linear average of the power intensities perceived on each subchannel during the sensing window. However, Abanto-Leon *et al.* [70] proposed a new nonlinear power averaging method, in which the most recent measurements are given a higher priority via exponential weighting. This approach is applied on both step 2 and step 3 of the SPS algorithm. The authors prove, through simulations, that their solution has shown a positive impact in terms of PRR performance, which enhances the reliability in V2V communication in mode 4.

Here, it is worth noting that a wrong resource selection can cause the packets loss problem for successive packet transmissions during the resource reservation period. For that, Bazzi *et al.* [71] defined this event as wireless blind spot (WBS) and tried to study the probability that it occurs. In order to avoid this problem, the authors propose an enhancement to the legacy SPS operation by limiting the maximum duration of any wrong reserved resource. In their proposal, each vehicle keeps an additional parameter that is used to define the maximum duration for the resource reservation period. The authors demonstrate through analytical results, validated by simulations, that this enhancement outperforms the legacy SPS algorithm.

Another important aspect related to the variation of the packet size has taken the attention of Molina-Masegosa and Gozalvez [72]. In fact, in V2V communication, the transmitted messages are formed by a first packet of 300 bytes followed by four packets of 190 bytes. So, let us assume, for example, that the packets of 300 bytes need two subchannels and those of 190 bytes need only one subchannel for transmission. In this case, just one of the two subchannels selected for the transmission of the packets of 300 bytes, which is reserved during the reselection counter, will be used for the transmission of the packet of 190 bytes. To overcome this problem, the authors propose that the resources are reserved using the reselection counter only for packets of 190 bytes, which are the more frequent ones. Whereas for packet of 300 bytes, the SPS is only used to select the adequate resource but this selected resource is not reserved during the reselection counter. The new proposed SPS achieves a better performance than the standard SPS algorithm in terms of PDR. Here, we note that the authors do not consider the case of aperiodic traffic such as the decentralized environmental notification messages (DENM).

Finally, the third category aims to propose new alternative solutions to mode 4 that are not based on incremental improvements to the SPS algorithm. In this context, Sabeeh *et al.* [73] proposed a new algorithm named estimation and reservation RA (ERRA) as a new alternative to the SPS algorithm. The ERRA algorithm tries to enhance the reliability of mode 4 by solving the collision problem and enhancing the PRR.

The main idea of the ERRA algorithm is that each vehicle lists all resource locations of the received packets that have the same random counter. With this solution, each vehicle can then predict the resources that will be free in the next resource selection. As a result, the vehicle does not need to depend on the sensing process. The ERRA algorithm is validated through simulations and the authors demonstrate that it presents a significant improvement compared to the standardized SPS algorithm. Then, the same authors have extended the ERRA algorithm to a new version, named extended-ERRA (E-ERRA) [74]. The main goal of the E-ERRA algorithm is to solve the problem of the lost reserved resources. This problem appears when a vehicle with preallocated resource has just left or appeared in the broadcast range before the reselection process. To solve this problem, E-ERRA specifies alternative resources that will be used by the vehicle if this problem appears. The authors prove through simulations that the E-ERRA reduces the collision ratio and increases the PRR. In the same context, Yang *et al.* [75] proposed a new algorithm for mode 4 named predictive assessment of resource usage (PRESS). In this algorithm, the transmitting vehicle should first predict the status of the channel for its future transmissions using the aggregate reselection counter information. By exploiting the periodicity of CAM messages, the PRESS algorithm can estimate the status of future resource usage at the time of resource use. With this solution, a vehicle can choose the least used resources. The authors demonstrate through simulations that the PRESS algorithm outperforms the standard SPS algorithm in terms of PRR.

In addition, Zhao *et al.* [76] proposed a new autonomous cluster-based resource selection scheme, which divides resource pool into orthogonal resource sets in order to reduce the probability of collisions. In their approach, the vehicles are grouped into clusters. Then, each cluster head (CH) selects a resource set that suffers less interference based on sensing. The authors prove that this approach, compared to the SPS algorithm, provides better performance in terms of PRR. However, the limitation of this work is that the selection of resource sets by the CH method may cause collision problems. The collision problem may occur when a CH selects a resource set that has already been selected by another CH outside its sensing range, which may affect the performance of the autonomous mode 4. Another interesting RA approach is presented in [77] that aims to mitigate the hidden node problem. The authors propose a new RA scheme aiming to reduce packet collisions. In this solution, vehicles select their resources based on the position and order of neighboring vehicles on the road. The proposed solution is evaluated analytically and through simulations and it is demonstrated that this new scheme outperforms the standard SPS algorithm. Despite the performance achieved by the proposed RA scheme, performance may be degraded if the positioning of vehicles, on which the proposed scheme is based, is inaccurate.

The above-mentioned proposed algorithms are globally focusing on the out-of-coverage areas in general. However, Sahin and Boban [78] focused on the RA in the delimited out-of-coverage area (DOCA). DOCA can be considered as a tunnel where no reception from the BSs is possible but the

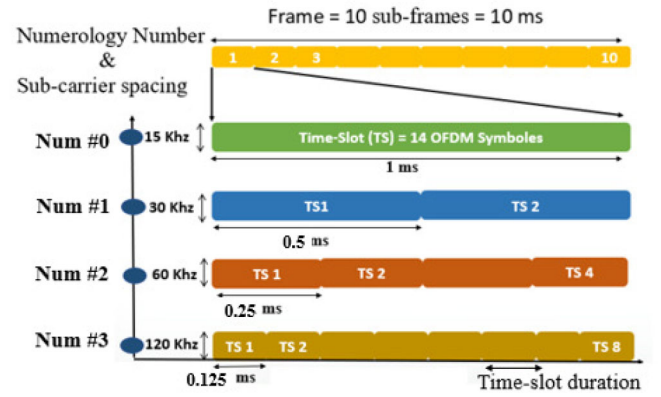


Fig. 9. NR-V2X frame structure.

BSs deployed at each end of DOCA are able to serve the vehicles just before their entry or after their exit. Their RA algorithm aims to improve the reliability for V2V communication. Therefore, the authors propose that the RA for the out-of-coverage vehicles entering the tunnel is still managed by the eNB based on the predictions of vehicle locations and the propagation conditions. The eNBs deployed at each end of the tunnel are able to serve vehicles just before they enter.

Despite the fact that mode 4 has been of interest of many researchers in recent years, there is still a need to further investigate other aspects related to collision avoidance, interference management, and congestion control mechanisms. So, new proposals are needed because this mode is considered as the baseline mode of C-V2X technology due to its independence of the cellular infrastructure. We summarize the state of the art of the main proposals for RA solutions in mode 3 and mode 4 in Tables IV and V, respectively.

IV. SIDELINK RESOURCE ALLOCATION IN NR-V2X

In this section, we first explain the general concepts related to the resources in NR-V2X. Second, we present, in-depth, the RA modes in NR-V2X, i.e., modes 1 and 2. Finally, we present the state of the art of the research works interested in modes 1 and 2 and the main issues discussed by the 3GPP members related to the potential future RA submodes.

A. Resources in NR-V2X

In NR-V2X, we still find the same radio frame of 10 ms and the subframe of 1 ms as in LTE-V2X [23] whatever the used numerology, as shown in Fig. 9. A physical resource block (PRB) is defined by 12 consecutive subcarriers in the frequency domain and a TS is defined by 14 OFDM symbols. The resource element (RE) is the smallest resource unit. It is defined in the frequency domain as well as in the time domain. An RE is defined by one subcarrier in the frequency domain for one OFDM symbol duration in the time domain. The duration of a TS depends on the used numerology. The TS follows the rule: $TS = 1/2^n$, where n indicates the order of the numerology. For example, in case of numerology 0, the TS value is 1 ms and it is of 0.5 ms for numerology 1. In Table VI, we present the parameters related to each numerology.

TABLE IV
STATE OF THE ART OF MAIN RA PROPOSALS IN MODE 3

Reference	Objectif	Scenario	Main results
[51]	Guarantee the Quality-of-Service (QoS) for each vehicle with avoidance of the main conflicts related to the intra-cluster and inter-cluster interference problems	Not defined	Achieve of the QoS requirements in terms of number of sub-channels and data rate needed by each vehicle.
[52]	Avoidance of the principle conflicts cited in [51] and enabling the reuse of resources between vehicles.	High vehicle density	High performance level in terms of the achieved data rate requested by vehicles.
[54]	Find the optimal one-to-one vertex assignment between vehicles and sub-channels	Not defined	Maximization of the capacity of the system defined as the achieved data rate.
[55]	Minimization of the overall power perceived by vehicles and the maximization of the resource reuse distance	Freeway scenario with 600 vehicles	Better performance in terms of the Packet Reception Ratio (PRR).
[56]	New resource allocation algorithm based on the geographical positions of vehicles.	A realistic and congested highway scenario, 2000 vehicles at 50 km/h [79]	Low appreciated error rate when using the GNSS positioning system compared to the UTD OA technique.
[57]	Maximization of the resource reuse distance between vehicles in order to control the probability of interferences	- Highway scenario (3 lanes per direction and linear density of 119 vehicles/km) - Urban scenario (667 average vehicles) [80] [79]	High performance level in terms of packet reception ratio and latency compared to [56].
[59]	New resource allocation solution based on the use of the geographical position of vehicles, the channel load and the traffic density	Urban and highway scenarios with 120 vehicles/km and 70 km/h [81].	Better performance than the existing LTE-V2X mode 3 and mode 4 scheduling schemes
[60]	Proposition of PEARL algorithm that aims to guarantee the QoS requirements of the V2X safety and non-safety applications	Highway scenario	High reliability and low latency
[61]	Extension of PEARL algorithm to PEARL-RM algorithm by adding new resource reuse mechanism	Highway scenario	Better performance in terms of reliability, latency and capacity utilisation.
[62]	New MIRD algorithm that aims to facilitate the reuse of resources between clusters of vehicles	Urban Scenario	Better performances in terms of PRR and low complexity

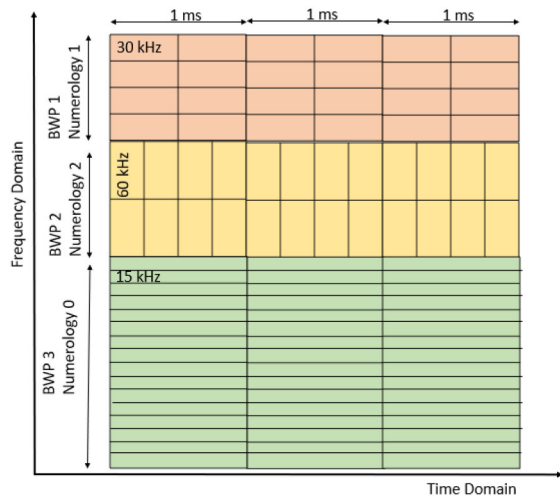


Fig. 10. FDM numerology multiplexing.

The multiplexing of different numerologies is another level of flexibility of the NR frame structure. The multiplexing of numerologies can meet the requirements of different applications. For example, due to their low-latency requirements, road safety applications can use a short TS. However, the infotainment applications require a high throughput, which can be ensured by a long TS.

The NR-V2X numerology multiplexing can be done either by the frequency-division multiplexing (FDM) technique as shown in Fig. 10, the time-division multiplexing (TDM) technique as illustrated in Fig. 11, or by mixing both FDM and

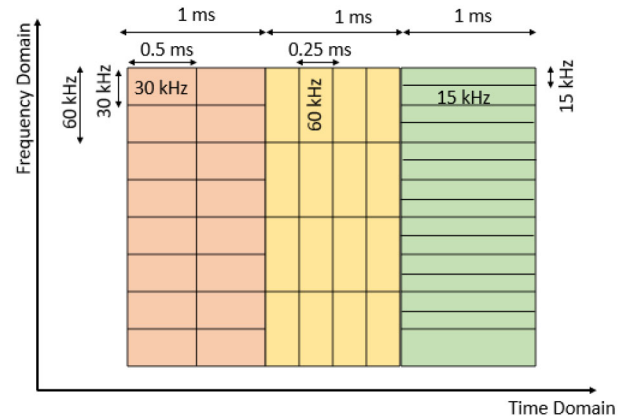


Fig. 11. TDM numerology multiplexing.

TDM techniques as presented in Fig. 12 [84]. The FDM-based numerology multiplexing is enabled by the bandwidth part technique (BWP) [23]. The BWP is introduced in 3GPP Release 15 and adopted for the NR V2X sidelink to support UEs that are not able to support large bandwidths, e.g., due to their processing or power consumption limitations. The bandwidth part, which is a contiguous set of PRBs on a given carrier, is defined for a particular numerology and may have the following parameters:

- 1) SCS;
- 2) CP length;
- 3) symbol duration.

TABLE V
STATE OF THE ART OF MAIN RA PROPOSALS IN MODE 4

Reference	Objective	Simulation scenario	Main results
[49]	Evaluation of the SPS algorithm	Urban scenario [81], 85 vehicles/km and 15 km/h	The SPS algorithm experiences significant packet collisions.
[63]	Study of the impact of the Resource Reservation Interval (RRI) on the performance of the SPS algorithm	Highway scenario [81], 70 km/h and 140 km/h	The Packet Delivery Ratio (PDR) increases with the increase of the RRI.
[64]	Study of the impact of the probability of resource reservation P and the RSRI threshold on the performance of SPS algorithm	Highway slow [81], 120 vehicles/km and 70 km/h	- The influence of P on the performance of the SPS algorithm depends on the channel load conditions. - Low values of RSRP threshold in step 2 of the SPS algorithm improve the performance of C-V2X Mode 4.
[47]	Evaluation of the performance of the SPS algorithm with analytical models	Highway scenario: 5 km with 4 lanes (2 lanes per direction) and speed of 70 km/h.	The performances of mode 4 are accurately represented by the analytical models.
[66]	Study of the impact of the non-periodical generation of the CAM packet on the overall performance of mode 4.	Highway scenario	The overall performance reduces as soon as the CAM packet generation interval is not equal to the allocation period.
[67]	Proposition of a new Counter Learning and Re-selection (CLR) mechanism.	Several scenarios: not detailed	Decrease of the probability of collision problems.
[68]	Proposition of a new reservation scheme in order to decrease the probability of collisions.	Urban scenario	High performance in terms of reliability and latency
[69]	Proposition of a new short-term sensing scheme before the resource selection phase	Highway Scenario [82]	Reduction of the packet collisions.
[70]	New non-linear power averaging method applied in the SPS algorithm	Urban and Highway scenarios	Better system performance in terms of Packet Reception Ratio (PRR) for both scenarios
[71]	New solution that aims to enhance the SPS algorithm by avoiding the Wireless Blind Spot (WBS) problem	Highway Scenario	The proposed solution outperforms the legacy SPS algorithm
[72]	New resource reservation procedure taking into account the different packets size of transmitted messages	Highway scenarios [81]	Better performance than the standard SPS in terms of PDR.
[73]	Proposition of ERRA algorithm that aims to enhance the reliability of mode 4	Different traffic scenarios	The ERRA presents a significant improvement compared to the standardized SPS algorithm.
[74]	Extension of ERRA to E-ERRA algorithm that aims to solve the problem of the lost reserved resources.	Highway scenario	E-ERRA reduces the collision ratio and increases the PRR
[75]	Proposition of PRESS algorithm that aims to enhance the sensing and the resource selection for mode 4	Highway scenario	PRESS algorithm outperforms the legacy SPS algorithm
[76]	New autonomous resource selection scheme based on the division of resource pool into orthogonal resource sets.	Urban and Highway scenarios [83]	Better performance in terms of PRR
[77]	New autonomous resource selection scheme based on the geographical positions of vehicles and the order of neighboring vehicles on the road	Highway scenario [81]	Enhancement of the performance of C-V2X communications.
[78]	Proposition of a resource allocation algorithm dedicated for the Delimited Out-of-Coverage Area (DOCA)	Highway scenario	Enhanced reliability for V2V communication in DOCA

TABLE VI
MAIN PARAMETERS OF NR-V2X NUMEROLOGIES

Numerology	Slots per sub-frame	Symbol length (μ s)	Slot length (ms)	Sub-carrier spacing (kHz)	Symbols per slot
0	1	71.42	1	15	14
1	2	35.71	0.5	30	14
2	4	17.85	0.25	60	14
3	8	8.92	0.125	120	14
4	16	4.46	0.0625	240	14

B. Sidelink Resource Allocation Modes in NR-V2X

As in LTE-V2X, 5G-V2X supports two RA modes. The first mode is the undercoverage mode known as mode 1 [85] and the second mode is the out-of-coverage mode known as mode 2 [86].

The 3GPP has discussed the resource pool sharing between NR-V2X mode 1 and mode 2. It is stated that these two modes can use separate resource pools, or they can share the same resource pool. Sharing the resource pool between mode 1 and mode 2 allows the more efficient use of radio resources.

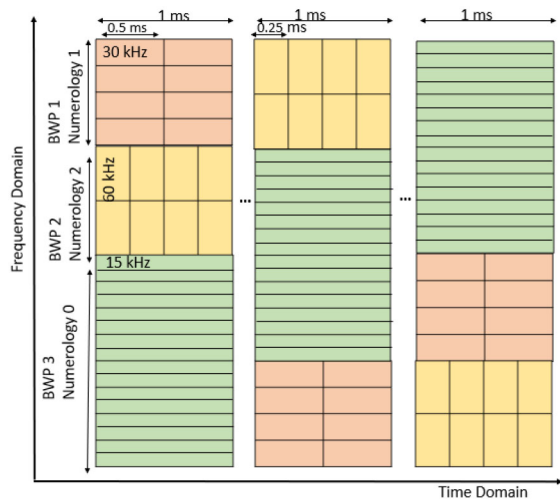


Fig. 12. Mix of TDM and FDM numerology multiplexing.

However, this may lead to collisions between vehicles operating in different modes. To resolve this problem, vehicles operating in mode 1 notify vehicles operating in mode 2 of the resources allocated for their future transmissions by using the SCI field.

1) *Mode 1*: In mode 1, the Next gNB schedules and allocates resources to vehicles. Like the LTE-V2X mode 3, the NR-V2X mode 1 uses the dynamic grant (DG) scheduling, but it uses the configured grant (CG) scheduling instead of the SPS used in LTE-V2X mode 3 [50]. In the DG, the vehicle requests resources to the gNB for each transmission using the physical uplink control channel (PUCCH). Then, the gNB responds with the downlink control information (DCI) over the physical downlink control channel (PDCCH). The DCI indicates the subchannels and slot allocated to the vehicle for the transmission of its message. Then, the vehicle informs other vehicles about the scheduled resources using the SCI. As a result, the nearby vehicles operating in mode 2 are informed about the resources that will be used by vehicles in mode 1.

The main drawback of the DG is the increase of the delay and the signaling overhead. For that, mode 1 includes the CG scheduling as a second option. In the CG scheduling, the gNB can assign a set of resources, referred to as a CG, to a vehicle in order to transmit several messages. To this aim, the vehicle sends a message to the gNB indicating information about the periodicity of the message and its maximum size. Also, it indicates the QoS information, such as the latency and reliability, required by the transmitted message. Thus, based on this information, the gNB creates, configures, and allocates a CG to the vehicle that meets the requirements of the sidelink traffic.

2) *Mode 2*: In mode 2, vehicles choose autonomously their appropriate resources. The resources in mode 2 are chosen from the sidelink resources configured by the gNB or pre-configured in the vehicles. However, in order to accommodate new use cases of the NR-V2X, 3GPP members have discussed some solutions for the improvement of the SPS algorithm used in mode 4 of LTE-V2X [87]. For advanced safety service,

NR-V2X supports the transmission of periodic and nonperiodic traffic with variable packet sizes, unlike LTE-V2X, which considers only the periodic traffic with fixed packet sizes. The 3GPP offers, in NR-V2X, a new channel sensing method, called short-term sensing, similar to the listen before talk (LBT) mechanism, as discussed in [88]. For mode 2, the proposed RA algorithm is the sensing-based SPS algorithm, already standardized for LTE-V2X mode 4, but the 3GPP proposals suggest to modify the sensing and the resource selection procedures in order to adapt to the periodic and the nonperiodic traffic. One of the most discussed open issue in mode 2 is the sensing and the resource selection procedures. There are two proposed channel sensing mechanisms.

- 1) *Long-Term Sensing*: The sensing used in LTE-V2X mode 4 is an example of the long-term sensing procedure. The main advantage of the long-term sensing consists of avoiding interference problems caused by the transmission with resources used by other UEs.
- 2) *Short-Term Sensing*: The short-term sensing is based on listening the medium before the transmission. The short-term sensing is more beneficial in case of nonperiodic traffic and it can be used as a complementary solution for the long-term sensing in order to avoid collision problems.

Many discussions have suggested that the long-term sensing is suitable for the periodic traffic and the short-term sensing is more adequate for the nonperiodic traffic. In fact, in case of nonperiodic traffic, the transmission behavior of a vehicle is not predictable, so the resource selection in LTE V2X mode 4 based on the long-term sensing is not applicable any more. The long-term sensing procedure used in LTE-V2X mode 4 can be reused in NR-V2X mode 2 for periodic traffic, where the short-term sensing based on LBT should be used for nonperiodic traffic.

The short-term sensing can be based on both the decoding of the SCI, as defined in Section III-A, and the sidelink measurement. In fact, the SCI is used to predict the channel resources occupation status. However, for nonperiodical messages, it can be difficult for the SCI-based decoding scheme to avoid potential resource conflict. In this case, the power measurement can be used to provide some reference information for resource selection.

The combination of the long-term sensing and the short-term sensing is also another solution. In this case, the long-term sensing can be used to determine and exclude occupied resources and form list of candidate resources for transmission within sidelink resource SW. Then, the short-term sensing operates within resource SW taking into account information from the long-term sensing procedure, i.e., a list of candidate resources, in order to select a particular resource for transmission within resource SW.

Here, we note that such proposals need to be tested in realistic scenarios. Thus, the choice of the appropriate sensing procedure will relatively depend on the validated experimental results. As the NR-V2X modes are recently defined by 3GPP, many research works are needed, in this context, to investigate more aspects related to the operation of these modes.

C. State of the Art on Resource Allocation in NR-V2X

As the NR-V2X is recently standardized by the 3GPP, there exist few research works, in the literature, investigating the RA in NR-V2X technology. In this section, we present the most recent research works interested in this subject. We first begin with the contributions interested in mode 1, then we present those related to mode 2. In addition, we present the research works studying the impact of the RA algorithms on the V2X applications, and especially the platooning application. Moreover, we also present new RA algorithms based on the machine learning (ML) technology.

1) *Mode 1*: Xiaoqin *et al.* [89] proposed a new RA algorithm for mode 1. In this work, the authors suppose that the gNB allocates resources to vehicles based on the reported channel state information (CSI). However, the reporting of the CSI can lead to the overhead problem. For that, in order to cope with this problem, the authors propose in a first stage a new power allocation scheme of CSI transmission, that aims to reduce the energy consumption in transmitting the CSI from the vehicles to the gNB. Then, in the second stage, the authors try to maximize the system throughput by modeling the sidelink radio RA problem as a mixed binary integer nonlinear programming (MBINP). However, we note that the authors of this article cited above do not consider the impact of high vehicle mobility on their proposed RA algorithm. Indeed, the resource scheduling based on the perceived CSI is not easy to implement in the vehicular context due to the high mobility of vehicles, which leads to the rapid change of the channel conditions.

Abbas *et al.* [90] proposed a two-level RA scheme that aims to reduce the latency, increase the throughput, and guarantee the reliability for V2V and V2N communications. In this work, the authors consider the scenario where the direct link between the gNB and a transmitting vehicle is blocked by a bypassing trailer. So, in this case, another vehicle can act as a relay between the gNB and the transmitting vehicle. In this scenario, the vehicle considered as a relay should obviously have a good connection with both the transmitting vehicle and the gNB. This vehicle can forward an uplink signal with the V2N link or a downlink one with the V2V link. For that, the authors consider two types of traffic, which are the V2V and the V2N communications, sharing the same radio resources. In this context, the authors propose an efficient RA algorithm that aims to avoid the interferences between the V2V and V2N vehicles and guarantee the different QoS requirements of both V2N and V2V communications. The simulation results show that this proposal enables to increase the throughput and decrease the latency. Another deep neural network (DNN)-based RA algorithm for mode 1 is proposed by Gao *et al.* The main goal of this algorithm is to maximize the system throughput by assigning to vehicles the optimal transmit power [91].

2) *Mode 2*: In the literature, there exist recent research works interested in the sensing-based SPS algorithm standardized for the autonomous mode 2 of NR-V2X. Among those research works, we cite the work in [92], where Ali *et al.* evaluated the impact of the main parameters of the SPS algorithm

on the overall system performance. In this work, the authors take into account different parameters, such as the NR flexible numerologies, the maximum number of resources per reservation, the MCS level, and the resource SW. However, Romeo *et al.* [93] focused on the reliability of the transmission of the DENM in this autonomous mode. In fact, the DENM, which are of aperiodic nature, are transmitted by a vehicle in order to prevent other vehicles of road hazard events. In this context, the authors are interested in the case where both the periodic and the aperiodic messages are transmitted in the same scenario and they share the same radio resources. They analyze the impact of the main parameters of the SPS algorithm of mode 2 on the DENM delivery ratio. They also study the impact of the generation of the aperiodic DENM messages on the reliability of the transmission of the periodic CAM messages. Furthermore, the authors study the effectiveness of using a short-term sensing with a duration of 100 ms for the transmission of DENM messages. They demonstrate that the use of a short-term sensing with multiple repetitions of DENM messages has an effective advantage on the DENM delivery ratio. However, the reliability of the CAM messages is affected when the load of the DENM messages increases in the scenario. In addition, Yoon and Kim [94] tried to enhance the standard SPS algorithm by proposing a stochastic reservation scheme for the aperiodic traffic. The main idea of their proposal is to predict the next transmission time of the next packet in order to reserve resources for this transmission. This prediction is done by analyzing the last packets intervals. The authors demonstrate through simulations that the proposed scheme achieves high PRR values. Most of the above-mentioned proposals do not consider congested scenarios, where the RA is more challenging due to the possibility of packet collisions and interference problems. However, Yi *et al.* [95] are interested in the congested intersection areas. For that they propose a novel sensing-based semi-persistent scheduling method in order to efficiently allocate resources to vehicles using the sensing information generated by the RSUs. In this work, the authors propose that the RSU acts as a central controller and periodically receives the CSI from each vehicle. Then, based on these informations, the RSU generates RSRP and S-RSSI maps that will be broadcasted to vehicles in order to better autonomously choose their radio resources. This proposed scheme enables to avoid the collision problems and so outperforms the standard SPS algorithm proposed for mode 2.

Another important aspect is also recently investigated in the literature, which is the study of the impact of the NR flexible numerologies on the performance of the autonomous mode [96], [97]. Ali *et al.* [96] focused on the study of the impact of the NR numerologies on the sensing and the SWs of the SPS algorithm. The authors demonstrate that when the resource SW size is the same for all numerologies, a higher SCS brings better performance and it is beneficial for the reliability of the basic safety service. In addition, Campolo *et al.* [97] evaluated the impact of the SCS on the reliability of the transmission of CAM messages in the autonomous mode. This work takes into account different conditions with different CAM periodicity values, channel

TABLE VII
STATE OF THE ART OF RA MODES IN NR-V2X

Reference	Mode	Objective	Simulation scenario	Main results
[89]	Mode 1	A new resource allocation algorithm that aims to: - Reduce the overhead caused by the reporting of the SCI by the vehicles to the gNB - Maximize the overall throughput for NR-V2X communications	Urban scenario	Low energy consumption in transmitting the CSI information and high system throughput
[90]	Mode 1	New resource allocation algorithm that aims to guarantee the QoS requirements for both V2V and V2N vehicles	Highway scenario [104]	Minimization of the probability of interferences between V2V and V2N vehicles with high throughput and low latency
[93]	Mode 2	Study of the impact of the autonomous mode 2 on the transmission of the aperiodic DENM messages	Not defined	The application of a short-term sensing increases the DENM delivery ratio
[94]	Mode 2	A new stochastic reservation scheme for the aperiodic traffic	Highway scenario of 3 km length	High performance in terms of Packet Reception Ratio
[95]	Mode 2	Proposition of a novel sensing-based semi persistent scheduling scheme in the congested intersection areas	Urban scenario with 2 lanes in each direction	Reduction of the probability of collision
[96]	Mode 2	Study of the impact of the NR numerologies on the performance of mode 2	Highway scenario	Higher Sub-Carrier Spacing (SCS) is beneficial for the basic safety service in autonomous mode 2
[100]	Mode 2	Study of the impact of the SPS algorithm on the platooning application	Highway Scenario	The impairments of the SPS algorithm reduce the performance required by the platooning application
[101]	Mode 2	Proposition of a new resource management scheme in the multi-cell scenario	Freeway scenario [105]	Increase of the system performance
[99, 106]	Mode 2	Proposition of a new deep reinforcement learning-based resource allocation scheme for platooning scenario	Manhattans Scenario defined in 3GPP TR 36:885 [107]	Minimization of the interferences with guarantee of the reliability and latency required by the platooning
[91]	Mode 1	New Deep Neural Network-based transmit power allocation	Not defined	Maximization of the overall system throughput of the V2X links
[102]	Mode 2	New Multi-Agent Reinforcement Learning-based resource allocation algorithm for platooning scenario	Urban scenario	Reduction of the Age-of-Information parameter and increase of the probability of transmission of the CAM messages.
[103]	Mode 2	New deep reinforcement learning-based resource allocation algorithm in the platooning scenario	Highway scenario	Decrease of the collision probability between platoon members

density values, and MCS levels. The authors demonstrate that increasing the SCS significantly improves the reliability of CAM transmission by increasing the PRR. Furthermore, it also reduces the Update Delay, which refers to the time difference between two successive receptions of two decoded CAM messages. However, the limitation of this work is that the authors do not consider the delay spread of the channel and the vehicle speeds as important parameters. In fact, these two parameters have a direct impact on the overall system performance, due to the intercarrier interference (ICI) and the intersymbol interference (ISI) problems caused by the Doppler shift and the multipath fading problem, respectively, as demonstrated in [98].

Here, we note that new techniques, such as the ML technology, are recently used to propose new RA algorithms. In this context, Ye and Li [99] proposed a decentralized RA algorithm based on a deep reinforcement learning technique for V2V communications. In this proposal, the agent, which is considered the V2V connection, can make its own decision to choose the optimal radio resource based on the CSI. The goal of this proposal is to minimize the interferences between

vehicles and meet the reliability and latency required by the V2V communication.

In Table VII, we summarize the state of the art of the main proposed RA algorithms for modes 1 and 2 of NR-V2X.

D. Impact of Resource Allocation on V2X Applications

The majority of the research works found in the literature are interested in the performance evaluation of the proposed RA algorithms for C-V2X, while others have proposed new alternative solutions. However, there is a lack of research works interested in the evaluation of the direct impact of the RA algorithms on the 5G-V2X applications. In this context, a particular attention is given to the platooning application as a very critical 5G-V2X application in terms of latency and reliability. In fact, the platooning is based on the advanced control techniques that require low latency and high reliability in the delivery of information. For that, recently, some research works are interested in the evaluation of the impact of the proposed RA algorithms on the required performance of this application. As examples Segata *et al.* [100] demonstrated

that the impairments presented in the SPS algorithm can have a negative impact on the performance required by the platooning application. In addition, Hegde *et al.* [101] proposed a new RA algorithm for platooning application in case of multicell scenario. In this proposal, the resources allocated to the whole platoon are granted by the gNB to the platoon leader and then the platoon leader distributes the RA information to its members. The authors demonstrate that this proposal enhances the system performance and overcomes the cell border issues.

In addition, there are recent works that propose new RA algorithms for the platooning application using new technologies such as ML. Among those research works, we cite the work in [102], in which, Parvini *et al.* tried to achieve the required latency and reliability in the platooning application in a high mobility scenario. To achieve their goal, the authors propose a new RA approach based on the multiagent reinforcement learning (MARL) technique, in which each platoon leader acts as an agent and interacts with its environment. The goal of this proposal is to minimize the Age-of-Information (AoI) parameter and maximize the probability of transmission of the safety CAM messages. In the same subject, Cao and Yin [103] proposed a new RA algorithm using the deep reinforcement learning technique that aims to decrease the collision probability in the platooning scenario. The authors demonstrate through the Monte Carlo simulations that the proposed algorithm outperforms the random resource selection which is a simplified version of the legacy autonomous mode 2 algorithm.

V. CHALLENGES AND OPEN ISSUES

After a comprehensive review of the literature, we present in this section the main interesting ideas concluded from the literature review. In a first part, we present the challenges related to the RA in V2X communications. In the second part, we present the interesting open issues that should be more investigated in the future research works.

A. Challenges

According the reviewed state of the art on the RA algorithms for C-V2X, we can summary the most interesting challenges as follows.

1) *Challenges Related to the Autonomous Mode:* As already mentioned, in the RA for C-V2X, the challenging RA mode is the autonomous mode, i.e., mode 4 in LTE-V2X or mode 2 in 5G-V2X, because of its decentralized behavior. The literature review has detected many challenging aspects related to the standardized SPS algorithm and try to resolve them by proposing new alternative solutions. Despite that, new efficient proposals are still needed. In the following, we present the two interesting challenges for the autonomous RA mode.

a) *Collision problem:* The collision is one of the most stringent problem in the autonomous mode detected by the literature review. In fact, the autonomous mode is based on the sensing of the channel for the resource selection procedure. However, a wrong sensing can lead to the choice of a radio resource already used by another vehicle, which can lead to

packet collisions. The wrong sensing occurs when two vehicles are unable to detect each other. As a result, one vehicle may choose a radio resource that is already in use by another vehicle. Furthermore, the packet collision can increase in the congested scenario with high vehicle speeds. This is explained by the fact that when a vehicle selects a resource and moving at a high speed, this vehicle can quickly come within range of another vehicle using the same resource that was not originally in its sensing range and was obviously not considered during the resource selection phase of the SPS algorithm. For that, we highlight that, even if many existing proposals in the literature try to solve the collision problem, there is still a need to further investigate this challenging aspect in the future research works.

b) *HD limitation:* Among the research works in the literature review, a lot of proposals highlight the limitations of the HD nature of the sidelink PC5 interface. In fact, the C-V2X equipment operates under the HD mode. As a result, a vehicle cannot transmit and receive simultaneously. For this reason, when a vehicle is under transmission, it is not able to receive other messages from its neighbors, which is very dangerous for the road safety when the received messages are belonging to the safety applications. The problems caused by the HD limitation of C-V2X equipment can be easily mitigated by the e/gNB in the network-controlled mode by allocating time-orthogonal resources to neighboring vehicles. However, in the autonomous mode, the vehicle does not have a holistic perception of the network topology, so the impact of the HD is greater in this mode.

2) *Challenges Related to the Network-Controlled Mode:* We have already highlighted that the network-controlled mode is not very challenging thanks to the control of the e/gNB. For that, the literature review addressing this mode is more less than the research works interested in the autonomous mode. However, in our point of view, there still exit challenging issues that need to be more investigated by the research community in the future. Among them, we cite the interference management for C-V2X communications, which is complicated by the rapid change of the topology due to the high vehicle speeds. In addition, the radio RA based on the channel quality indicator (CQI), reported from the vehicles to the e/gNB, is also another challenging task. In fact, the high vehicle speed leads to the rapid change of the channel quality, for that it is challenging to allocate resources to vehicles based on their reported CQI. For these reasons, we note that those highlighted aspects should be taken into account as future research directions.

We highlight also another challenging aspect, namely, the efficient allocation of resources to vehicles in order to meet the requirements of different V2X applications. As already indicated in Section IV-D, there is a lack of research works investigating the impact of the RA algorithms on the requirements of different V2X applications. Among those applications, we cite the advanced cooperative driving as an example of a 5G-V2X time-critical application, which is part of the ultrareliable low-latency communications (URLLCs) and requiring a very low latency. In addition, we cite also the 5G-V2X infotainment applications such as the video streaming for passengers,

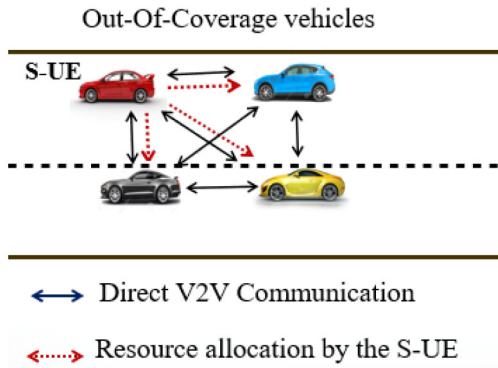


Fig. 13. Submode 2(d) of NR-V2X.

which is part of the enhanced mobile broadband (eMBB) and requires high data rates. So, due to the various requirements of V2X applications, many research works are needed to investigate this research subject, in order to significantly improve the scheduling process for 5G-V2X communications.

B. Open Issues

In this section, we present the open issues related to the RA in C-V2X, that need to be more investigated in the future.

1) *Open Issues Related to Future Resource Allocation Modes of 5G-V2X*: The 5G-V2X has standardized two modes of RA as already mentioned in Section IV-B. However, there are other potential submodes of mode 2 that were analyzed during Rel.16, but that were not standardized. Those submodes still considered as particular use cases of NR-V2X mode 2 and could be of interest in future 3GPP releases.

The first one is known as submode 2(b), in which a UE can assist the resource selection of another UE with the assistance information [108]. This assistance information can include several parameters such as the geographical position or the velocity of vehicle, the priority of traffic, the CR, or the RSRP. The submode 2(b) is discussed in Release 17, as presented in Section VI.

The second mode of RA, which is discussed in Release 16, but is not adopted in neither Release 16 nor Release 17, is the submode 2(d), in which, a vehicle known as scheduling UE (S-UE) is responsible for the scheduling of radio resources for a group of vehicles [109], as illustrated in Fig. 13. For this submode, one of the most important issues is how to choose the S-UE. The choice of the S-UE can be done considering, for example, the geographical position or the hardware capacity of vehicles. Here, we note several open issues about the operation of this submode. First, which resource the S-UE should use when allocating resources to its members? Second, is there a probability of interference between vehicles allocated by different S-UEs? Third, how can the S-UE indicate the allocated resources to its members and what is the procedure for members when the S-UE disappears? Finally, what is the adequate scenario for the operation of this submode?

In this context, we point out that the platooning scenario, obviously, seems to be the easiest scenario for the operation of the submode 2(d). The stability of the group of vehicles facilitates the RA task for the platoon leader. However, this

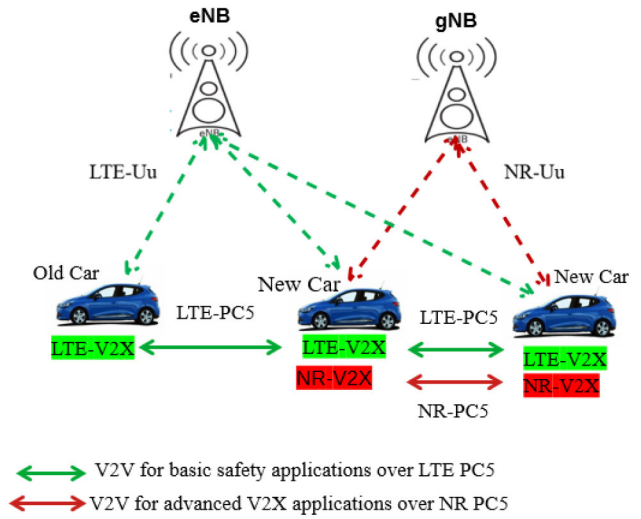


Fig. 14. eNB/gNB control of sidelink V2V resources.

assumption does not preclude the possibility to apply this submode for the other scenarios. We suggest to consider, for example, the position, speed, and direction of vehicles, as decisive parameters, in order to choose the stable scenarios, in which, the operation of the submode 2(d) is easier.

Finally, it is important to note other drawbacks of this submode. In fact this submode may not offer reasonable performance advantages because it may increase the latency and it generates extra signaling overhead. Additionally, this submode may cause interference problems among UEs belonging to different groups. In this context, we conclude that the submode 2(d) needs more research works and experimental results in the future in order to have a more in-depth vision of its advantages and its drawbacks.

2) *Other Potential Open Issues*: Last but not least, it is worth noting that there are some other topics of interest that are also investigated by the 3GPP. As discussed in [110], it is not intended for NR-V2X to replace the services offered by LTE-V2X. In contrast, the NR-V2X is intended to complement LTE-V2X for advanced V2X services and support interoperability with LTE-V2X. Thus, the possibility that an eNB controls the NR sidelink resources and the ability that a gNB controls the LTE sidelink resources are also technical challenges as discussed in [33] and [111], respectively. For instance, in the case that a vehicle is equipped with both RATs and it is out of the NR coverage but in the LTE coverage, the eNB can control the NR sidelink resource scheduling for advanced V2X applications through the LTE U_u interface. Similarly, when a vehicle is not in the LTE coverage, the gNB can control the LTE sidelink resource scheduling for the basic safety service, as presented in Fig. 14.

Finally, we identify other research issues that still need attention, such as the resource sharing between mode 1 and mode 2 and the switching from one mode to another.

VI. FUTURE TECHNICAL TRENDS AND POSSIBLE SOLUTIONS

In this section, we present the future attractive research directions related to the RA in C-V2X technology. First, we

present the RA enhancements that 3GPP is discussing for Release 17. Then, we discuss the application of the new emerging technologies in the RA subject, such as the ML, the mobile-edge computing (MEC), and the network slicing (NS) technologies.

A. Discussion of RA in 3GPP Release 17

Among the main topics in the 3GPP discussions in Release 17 on improving V2X sidelink communications, we can mention improving reliability, reducing latency for URLLC, and increasing security of V2X sidelink communications. However, we can summarize the specific improvements related to RA under the following aspects.

1) *Enhancement of Mode 2 for Power Consumption Reduction*: One of the most interesting RA improvements agreed upon for Release 17 is the improvement of NR sidelink mode 2, which is standardized in Release 16, by addressing the power consumption issue. 3GPP plans to adopt a new version of mode 2 in Release 17 that can reduce the power consumption of UE devices [112]. Indeed, the power consumption is not a major problem when considering UEs, such as vehicles or RSUs. However, this is the case when it comes to other types of UEs, such as UE devices used by pedestrians and cyclists for V2X sidelink communication. The autonomous sidelink mode 2, which is standardized in Release 16, is based on the sensing of the channel during a sensing window, which consumes a nonnegligible amount of the energy from the transmitting devices. To address this issue, 3GPP plans in Release 17 to replace the first channel sensing method with partial sensing and random resource selection schemes, which can significantly reduce the overall power consumption of UE devices. However, we note that it is interesting to study the performance of the new proposed methods compared to the traditional autonomous mode 2, which can be considered as an interesting future research direction.

2) *Inter-UE Coordination*: The assistance of one UE in resource selection by other UEs [112] is already known as mode 2(b) in Release 16 but was not standardized. The inter-UE coordination includes two types: type A or type B, where one UE restricts or suggests the resources that can be selected by another UE, respectively. One of the advantages of inter-UE coordination is that it avoids the hidden terminal problem, which may occur when a transmitting vehicle V_1 is not in the range of another vehicle V_3 and therefore does not know which resources are already being used by that vehicle. In this case, the transmitting vehicle V_1 can choose the same resource that is already used by vehicle V_3 , which can lead to packet collisions. For this purpose, an intermediate vehicle V_2 , which is in the range of both vehicles V_1 and V_3 , can recommend or restrict the resources to be used by vehicle V_1 .

B. Machine-Learning-Based RA

Nowadays, the ML or artificial intelligence (AI) is a new emerging technology that draws the attention of the scientific research community [113], [114]. This new emerging technology is widely applied in the wireless communication domain.

However, the application of the ML in the RA subject is considered as one of the most important future trends that need to be more investigated by the research community in the future. In fact, in the vehicular network context, there already exist some recent research works [99], [102], [103], [106] investigating the RA based on the ML technique. However, the V2X use cases are different from the cases in which ML is typically used. This is due to the fact that the vehicular networks have specific characteristics, such as the high mobility of vehicles, the rapid change of the topology, the multitude V2X use cases with different requirements, and more others. As a result, we highlight the interesting open issue in this context, which is how to efficiently allocate resources to vehicles based on historical data? For that, there is a need to more investigate the use of this new emerging technology in C-V2X communications.

C. RA for Vehicular Edge Computing

The MEC is one of the new 5G emerging technologies, recently used in very large-scale networks. MEC is beneficial for time-critical applications thanks to the storing and processing of contents and applications in the proximity of mobile users. In the vehicular context, in the near future, the connected vehicles are also expected to use the advanced onboard sensors, such as the cameras, radar, and lidar. Consequently, a large volume of data will be generated, processed, and transmitted with a very low-latency requirement. Therefore, MEC technology is considered a promising tool for analyzing such huge amounts of data to enhance the performance of vehicular networks. The utilization of MEC in vehicular networks is known as vehicular edge computing (VEC) and is considered an important research direction that has been widely investigated by the research community in recent years [115], [116]. However, the radio RA in VEC is generally associated with computational RA, as studied in [117]–[120]. The reduction of the total latency, which includes the transmission time at the radio interface and the processing time at the edge server, is one of the main goals in joint RA for VEC. To achieve this goal, the allocation of radio resources, which affects the transmission time, must be managed efficiently. In addition, allocating high computational resources is the best choice to reduce the processing time at the edge server. Therefore, the question of how radio resources can be allocated to reduce the transmission time and how many computing resources should be allocated at the edge server to reduce the processing time is an important research direction that needs to be investigated in more detail in the future.

D. Network Slicing-Based RA

The NS, enabled by the network function virtualization (NFV) technology, is one of the new emerging technologies of 5G enabling the building of multiple virtual networks on a common shared physical infrastructure [121]. The NS is crucial in the 5G network because of the different QoS requirements of the multitude services.

As the NS is a new emerging technology, there are few research works investigating its application in the vehicular

network context. In fact, the multitude use cases of vehicular applications do not enable the mapping of all V2X applications on only one slice of the three well-known slices, which are the eMMB slice, the massive machine-type communications (mMTCs) slice, and the URLLC slice. For this reason, it is expected that various V2X applications shall be mapped separately in different slices. As a result, there is a need to efficiently allocate resources to vehicles to meet the requirements of different V2X applications belonging to different slices. In this context, the assignment of different numerologies to different V2X applications is an example of the slicing of radio resources in the radio access network (RAN). As an example of a time-critical V2X use case, we cite the advanced and remote driving, which needs a large SCS to ensure the ultralow latency and high reliability required for this use case. Finally, we highlight that the slicing of radio resources forms a future research aspect, which needs to be more investigated as a very important attractive direction.

VII. CONCLUSION AND PERSPECTIVES

The C-V2X is a prominent enabler of the IoV paradigm. The RA in C-V2X has a major impact on the performance of V2X communications. In this work, we provide an extensive state of the art on RA in C-V2X technology. We present the existing research works related to modes 3 and 4 of LTE-V2X. We also present the most recent published research works interested in mode 1 and mode 2 of NR-V2X.

We conclude that the major challenge faced in LTE-V2X mode 3 is the small number of research works interested in this mode. On the other hand, several research works investigate the performance of mode 4. However, it still presents other critical issues related to collision, interference, and congestion control mechanisms.

For NR-V2X, as it is recently standardized by the 3GPP, there is a need to more investigate the RA subject and propose new RA algorithms that aim to enhance the 5G-V2X communications and meet the requirements of URLLC services.

In addition, we identify a number of research issues that still need attention, such as 1) the overall operation of the future submode 2(d); 2) resource sharing between mode 1 and mode 2; 3) switching from one mode to another; and 4) the control of the NR sidelink resources by the eNB or the control of the LTE sidelink resources by the gNB.

Finally, we present the future technical trends and possible solutions related to the RA in C-V2X using the new emerging technologies, such as the ML, the MEC, and the NS.

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